MARITIME AND SPACE LAW, COMPARISONS AND CONTRASTS (AN OCEANIC VIEW OF SPACE TRANSPORT)

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Welcome to the new age of outer space. President Carter has stated that the advent of the shuttle will mark the second era in outer space development.[1] It is very likely that new transportation systems will accelerate manned activity in this new domain at a geometric rate.

Few realized, when the first Echo satellites were launched in the early sixties, the dramatic way in which communications satellites would fill the skies.[2] Nor did they realize how crowded the geostationary orbit would become and how great the need would be for international regulation. The first satellite to operate in geostationary orbit was Syncom 2, which was launched in 1963. Between one hundred and ten and one hundred and twenty satellites now occupy that orbit. From one-quarter to one-half of them are functioning.[3] A recent NASA study concluded that by the year 2000 there would be a tenfold increase in the international demand for communications satellite circuits.[4]

The emergence of a transportation system in space, particularly a reusable system, will promote the same exponential growth in the carriage of cargo, and in human activity. How will it grow? Will it be by each nation regulating its own transport systems independently, by commonly agreed upon standards and practices, or by a new international agreement regulating space transportation?

The answer lies partially in the degree to which the most technologically advanced states believe it to their advantage to work out internationally accepted practices for their space operations. The U.S. will have the Space Shuttle, the USSR their Salyut-Soyuz-Progress system, and France has Ariane. The Soviets are reported to be planning a reusable spacecraft.[5] China and Japan are trying to develop more sophisticated launch systems.[6] The answer also lies in how much priority the U.N. Committee on Peaceful Uses of Outer Space (COPUOS) assigns to the formulation of an international regime for space transportation. The subject of space transportation systems was an agenda item at the most recent session of the Scientific and Technical Subcommittee, but there was

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[1] Press, "U.S. Space Policy—A Framework for the 1980's," 35 Astronautics and Aeronautics 34 (1979).

[2] *See* Lay and Taubenfeld, The Law Relating to Activities of Man in Space 109-121 (1970).

[3] This is the estimate of Dr. Lubos Perek, former Chief of the Outer Space Affairs Division.

[4] N.Y. Times, Mar. 24, 1980, § D, at 1.

[5] Aviation Week and Space Technology, Jan. 8, 1979, p. 11; Feb. 18, 1980, p. 25.

[6] See U.N. Secretariat Report, International Implications of New Space Transporation Systems 7, 8 (Aug. 16, 1979).

no substantive discussion. The subject was not on the agenda of the recent session of the Legal Subcommittee.

A current report prepared by the International Aeronautical Federation at the request of the Scientific and Technical Subcommittee states:

". . . it is conceivable that with the projected growth in both launches and the population and size of orbiting satellites there will come a time when the probability of interference with spacecraft performance, and possibly even physical collision, may become high enough to require consideration [of a global traffic control system]."

This, in my view, is most certainly an understatement. It is not only conceivable, but inevitable. The only issue subject to doubt is as to the timing.[7]

Whether a space transport regime takes shape through unilateral practice, concordant national rules, or multilateral treaty, the maritime and aviation regimes for international carriage are important references. Commerce at sea and in the air space has flourished by virtue of stable, internationally accepted rules, which regulate navigation. The purpose of this paper is to examine a few of the more important aspects of the maritime analogy for their applicability to space transport.

Status of the Instrumentality

None of the existing multilateral treaties regulating the use and exploration of outer space provide any definition of a spacecraft or a space vehicle. To my knowledge no bilateral treaty does. A recent amendment to the NASA Act defines *space vehicle* as "an object intended for launch, launched or assembled in outer space, including the Space Shuttle and other components of a space transportation system, together with related equipment, devices, components and parts."[8] This definition seems broad enough to cover all space objects contemplated by the Outer Space Treaty.

The terms *spacecraft* and *space vehicle* seem to be used interchangeably in the Outer Space Treaty and in the Astronaut Agreement, and frequently refer to space objects which carry personnel.[9] The Liability and Registration Conventions refer to neither spacecraft nor space vehicle, only space objects and launch vehicles.[10] A great deal of confusion exists as to the exact meaning of the two underscored terms. The former Chief of the Outer Space Affairs Division, Dr. Lubos Perek, has stated that in technical literature, the terms *spacecraft, space vehicle,* and *space object* all mean the

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[7]*Id.* at 18.

[8]Pub. L. No. 96-48, § 308 (f), 93 Stat. 349; 42 USC 2458 b.

[9]Treaty on Principles Governing the Activities of States in the Exploration and Use of Outer Space, Including the Moon and Celestial Bodies (hereinafter Outer Space Treaty), Jan. 27, 1967, [1967] 18 U.S.T. 2410, T.I.A.S. 6347, 610 U.N.T.S. 205, Art. 5 (effective Oct. 10, 1967); Agreement on the Rescue of Astronauts, the Return of Astronauts and the Return of Objects Launched into Outer Space (hereinafter Rescue and Return Agreement) April 22, 1968. [1969] 19 U.S.T. 7570, T.I.A.S. 6599, 672 U.N.T.S. 119, Art. 1-4 (effective Dec. 3, 1968).

[10]Convention on the International Liability for Damage Caused by Space Objects (hereinafter "Liability Convention") March 29, 1972 [1973] 24 U.S.T. 2389 T.I.A.S. 7762 (effective Oct. 9, 1973); Convention on Registration of Objects Launched into Outer Space (hereinafter "Registration Convention"), January 14, 1975, [1978] 28 U.S.T. 695, T.I.A.S. 8480 (effective Sept. 15, 1976).

same thing.[11] This is supported by the McGraw-Hill Dictionary of Scientific and Technical Terms. It defines *spacecraft* as "devices manned and unmanned which are designed to be placed into an orbit about the earth or into a trajectory to another celestial body." (No definition is given in the dictionary for *space vehicle*.)

There has always been a need accurately to define aircraft and ships and to distinguish them from other objects which transit or occupy their respective spheres. A similar need will emerge to define those spacecraft used for transport and to distinguish them from other space objects.

The International Regulations for Preventing Collisions at Sea define vessels to include, "every description of water craft, including non-displacement craft and seaplanes, used or capable of being used as a *means of transportation* on water"[12] [emphasis added]. The INMARSAT Agreement defines the term *ship* as meaning "a vessel of any type operating in the marine environment (including inter alia) hydrofoil boats, air-cushion vehicles, submersibles, floating craft and platforms not permanently moored."[13]

It seems that as space travel grows, we will need to establish a common definition of transport spacecraft or space vehicles. That is, spacecraft used primarily to carry goods or personnel from one place to another. It will also be necessary to distinguish them legally from other varieties of space objects. The key to defining the spacecraft for transport is navigability. Does it have the *primary* function of transportation? Is it designed basically for space flight rather than parking in a particular orbit? The fact that many satellites have internal rocket propulsion enabling ground controllers to reposition them in space does not endow them with navigability. Communications, remote sensing, weather, and reconnaissance satellites are not designed or intended for navigation in space. They carry only sufficient fuel to reach the desired orbit and perform their mission. INTELSAT satellites have been recently supplied just enough extra fuel to kick them out of the geostationary orbit at the end of their useful life. A true navigable craft however has enough fuel to transverse space and return to a particular place or orbit.

Navigate has been defined as steering or directing a ship or aircraft.[14] Of all the space objects, it will be the spacecraft for the transport that most nearly resembles ships and aircraft. How should they be described to differentiate them from other space objects? Webster's dictionary defines *spaceship* as "a rocket propelled vehicle for 'travel' in outer space."[15] One might have preferred a definition which substituted the word *transport* for the word *travel*. It is broader. Transport clearly embraces cargo as well as human beings. If one adopts the test of navigability as described above, then a spaceship could be manned or unmanned. The U.S. Space Shuttle clearly qualifies as a

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[11]Interview with Dr. Perek, New York, New York, March 27, 1980.

[12]Convention on the International Regulations for Preventing Collisions at Sea (Oct. 20, 1972) T.I.A.S. 8587, Rule 3.

[13]Convention on the International Maritime Satellite Organization, INMARSAT, July 7, 1979 [1979] T.I.A.S. 9605, Art. 1.

[14]Webster's New World Dictionary 948 (2nd ed. 1974).

[15]*Id.* at 1364.

spaceship. So, I submit, do the unmanned transport orbiters being developed by the Soviet Union, Japan, and the European Space Agency.

Once having defined the spacecraft for transport as a spaceship, its classification according to ownership and use becomes important. At sea, different rights and responsibilities flow from a ship's legal characterization. There are warships, other governmental vessels (with a subclassification depending on commercial or public use) and private merchant vessels. At the highest end of the scale is the warship. It enjoys complete sovereign immunity from foreign jurisdiction and has certain extraordinary rights on the high seas. Other governmental ships used for public purposes also have sovereign immunity, but not the special rights of warships. In most states, governmental ships used for private purposes no longer have immunity and, of course, neither do private merchant vessels. It has been necessary to classify vessels in order that all who use the high seas, or participate in maritime activity, understand the respective rights and duties of each type of vessel.[16]

The dawn of the new space transport era will bring a similar need to classify spaceships. The latest draft of the Law of the Sea negotiating text defines a warship as one bearing distinctive marks, commanded by a commissioned officer, in the service registry, and with a crew subject to military discipline.[17] A recent report indicates that out of a total of thirty-nine U.S. shuttle missions scheduled through September 1984, five are dedicated military missions.[18]

As the capabilities of the shuttle are demonstrated through experience, this is probably an underestimate of both the total number and the number of military missions. Should spaceships from the very outset be separately classified according to mission? In my view the answer is yes. Unlike other space objects, spaceships are instrumentalities of navigation. They are capable of a wide variety of tasks including military ones. Those ships operated by the military for military purposes, like their counterparts at sea and in the airspace, need particular identification. Following along the lines of the Law of the Sea draft, military spaceships should be defined as those which are under military control and whose ground controller or space crew are subject to military discipline.

At the initial stage of development, the spaceship will be an experimental vehicle, serving predominantly science, exploration and national security. As it begins to achieve economies that make it commercially useful for industry and for non-launch states, the spaceship will become an instrument of trade and commerce. It is interesting to note that under the U.S. Sovereign Immunities Act, a foreign state is not immune from U.S. jurisdiction in admiralty actions to enforce maritime liens against governmental vessels or cargoes used in commercial activity.[19] However, U.S. public vessels may not be

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[16]For a general discussion of the legal regime of warships and merchant ships, *see* Chapters VII and VIII, Colombos, International Law of the Sea 259, 285 (6th ed. 1957).

[17]Art. 29, Revised Informal Composite Negotiating Text for the Eighth Session, United Nations Conference on the Law of the Sea, March 19-27, 1979. The text is in 18 Int'l. Legal Materials 686, 709 (1979).

[18]Aviation Week and Space Technology, Mar. 31, 1980, pp. 54-55.

[19]Foreign Sovereign Immunities Act of 1976, 28 U.S.C. 1605(b). Under the Public Vessels Act, however, no lien may be created against any public vessel of the U.S. 46 U.S.C. 781, 788. In some foreign states, U.S. public vessels used for commercial activity may be subject to a libel action.

libeled in the United States. Some foreign nations refuse to accord any immunity to property owned by foreign governments and used in trade and when shuttle orbiters carry substantial commercial payloads, they may well be equated to public property used for private purposes and subject to attachment.

Dr. John Cobb Cooper, the first Director of the Institute of Air Law at McGill, wrote that vessels have a *sui generis* status.[20] While land transportation entities have no distinctive status, and aircraft have only nationality, vessels also possess a quasi personality which permit them to be held personally responsible for contracts and torts independent of owner responsibility. Justice Story wrote that a shipowner's innocence of any offense made no difference in the condemnation of a vessel suspected of "piratical aggressions."

"It is not an uncommon course in the admiralty, acting under the law of nations, to treat the vessel in which or by which, or by the master or crew thereof, a wrong or offense has been done as the offender, without regard whatsoever to the personal misconduct or responsibility of the owner thereof."[21]

Without delving into the subject of maritime liens, it is apparent that to vest spaceships with the same quality of responsibility attributed to vessels could have dramatic implications. A spaceship, part of whose payload is carried under contract with a private corporation, which lands on foreign soil through intent or accident, might find itself the subject of an in rem proceeding brought by a foreign claimant. Nothing in the Liability Convention prevents judicial recourse by private claimants even where that Convention might apply.

As spaceship operations become routine and the options they afford for commercial venture more advanced, their legal status becomes a matter of great importance and so will the analogy to public and private vessels.

The Right of Passage

Defining and classifying spaceships is only the start. Greater activity in space means more launches, which could mean more launch sites, or, as they will certainly come to be known, spaceports. There are probably fewer than a dozen separate launch facilities in the world today. Since the cancellation by Zaire of a contract with OTRAG, a private West German company for the development of a private launch facility in that country, all launch sites are owned and operated by Governments or Intergovernmental Organizations. However, there are those who believe that commercial launch sites are on the horizon and that they will multiply as commercial space ventures prove feasible. Eight nations have expressed interest in establishing new international launch facilities. Five of these are equatorial states. Will there be, before the end of the century, an explosion in space transport to equal that generated by air transport in the last quarter of a century? It is certainly a distinct possibility.

The rapid proliferation of communications satellites may prove an interesting precedent. Technological advances and economies of increased spacecraft production

[20]Cooper, The Legal Status of Flight Vehicles Explorations in Aerospace Law 206-213 (Vlasic ed. 1968).

[21]*The Malek Adhel*, 43 U.S. 210, 233 (2 How. 1844).

will provide direct access to space for more and more states. This will require the greater traverse of foreign airspace.

Many writers have addressed the problem of innocent passage through airspace to the outer space and have suggested the development of a rule comparable to the right of innocent passage for ships through territorial waters.[22] I prefer the analogy to the landlocked state. That state has no access to the unique environment of the oceans except through the territory of another. It has no sea coast. Without a freedom of transit such a state cannot reap the benefits of ocean transport. In the same manner, states who develop the capability to launch vehicles in space may not be geographically positioned to do so without transiting foreign airspace. They too are landlocked. The Geneva Convention on The High Seas provides that landlocked states shall enjoy the right of free access to the sea and equal treatment in ports.[23] A similar right is needed in space. It is at the outset of an active transport era that the greatest opportunity lies for the establishment of the principle of free transit to and from space. This right is less likely to emerge from the practice of states than by bilateral or multilateral agreement. Free transit through foreign air space would reduce the urgency to set forth a geometric border between air and outer space.

The Exercise of Exceptional Jurisdiction

The maneuverability of the spaceship will make it possible to approach and inspect alien satellites which are suspected of being a threat to the earth environment. It will be possible to remove the satellite to a high energy orbit, neutralize it, or control its return to earth.[24] The Outer Space Treaty makes it clear, however, that the registry state retains jurisdiction and control over its space objects and undoubtedly this is meant to be exclusive. Any exercise of control over a foreign satellite would have to be on exceptional grounds.

The corollary to the principle of the freedom of the high seas is that all states normally have a duty not to interfere with foreign ships outside territorial waters. In a famous admiralty case a British Judge declared,

> In places where no local authority exists, where the subjects of all States meet upon a footing of entire equality and independence, no one State, or any of its subjects, has a right to assume or exercise authority over the subjects of another.[25]

However, there is an exception to this principle of freedom of navigation. This exception involves the high seas right of approach. It is the right of any warship to approach a suspicious merchant vessel and verify her nationality and ensure she is not

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[22]*See* e.g., Christol, "Innocent Passage in the International Law of Outer Space," 7 Jag L. Rev. 22, 29 (1965).

[23]Convention on the High Seas (1958) T.I.A.S. 5200, Art. 3.

[24]*See* this writer's view on the removal of earth threatening satellites in 3 Ann. Air & Space L. 383-394 (1978).

[25]Le Louis 210 (2 Dods. 1817) quoted in Colombos, *supra* n. 16, at 310.

violating international law relating to fishing, submarine cables, and piracy, as well as misuse of the national flag. If the crew of the warship decides to board the suspect vessel, there is a heavy burden to substantiate the suspicion. Failure to do so makes the flag state liable for the loss, damage or delay to the merchant ship.[26] Whether a similar exceptional right of approach will develop in space rests on the good will of spacefaring states. Both the fall to earth of the Soviet Cosmos 954, breaking up over an area of northwest Canada about the size of Austria, and the uncontrolled return to earth of parts of the eighty-five ton Skylab, illustrate the risks involved in deteriorating satellites.[27]

The use of nuclear power sources in space is now being studied by a working group of the Scientific and Technical Subcommittee.[28] But nuclear power is only one specific source for earth contamination. There are other threats as well, including the possibility of physical damage. The ultimate answer lies in an international agreement for the safe deorbit or reorbit of satellites which threaten the environmental norm. Until such agreement is reached, military spaceships of any nation should have the right to approach a hazardous satellite and take preventive measures.

Safety of Navigation

It might appear at present that space voyages will not require the type of uniform rules for navigation that govern at sea. Space is limitless, the public oceans are not. Sea transit will far exceed space transit for decades to come. The risk of actual collision between two manned craft in space is minimal when compared to shipwreck and collision. However, the near earth and geostationary orbits will be the most frequented areas of space and as noted above there is already overcrowding in the latter orbit. As fully reusable launch systems transport large payloads from earth to low orbit and return them, and as reusable space-based orbital vehicles develop to transfer men and equipment from low to high orbit and to land on celestial bodies, uniform rules will be required to govern landing, docking, staging and inter-orbital transfers. It is an inexorable certainty that a navigation code will become as necessary for space safety as it became for traffic safety in the air space and at sea.

International rules of navigation applicable to high seas navigation have been enacted by virtually all maritime states, and amendments are usually made simultaneously by international agreement. The International Regulations for Preventing Collisions at Sea of 1972 are a part of these rules. They set forth measures to avoid collision, for maneuvering, towing, warning and distress signals.[29] There have been a number of conferences between maritime states called for the purpose of

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[26]*Supra* n. 16, at 311.

[27]For an excellent analysis of the deorbit of Cosmos 954, see Galloway, "Nuclear Powered Satellites: The U.S.S.R. Cosmos 954 and the Canadian Claims," 12 Akron L. Rev. 401 (1979).

[28]*See* Report of the Scientific and Technical Subcommittee on the Work of its Seventeenth Session, A/AC.105/267, Annex II, 1 (1980).

[29]*Supra* n. 12, rules 12-33.

standardizing rules for structural safety, fire and life saving equipment, and the use of radio and other navigational aids.

Whether an international space traffic code emerges through the concordant national legislation and regulation of the principal spacefaring nations, or through formal international agreement, it will become imperative to achieve uniform standards and practices. The Scientific and Technical Subcommittee working group on the use of nuclear power sources should be invited to consider the larger question of spaceship safety and to set forth suggested rules for uniform practices.

Status of Crew and Others

The Outer Space Treaty provides that all parties should regard astronauts as envoys of mankind.[30] What special protection this affords, apart from providing all possible assistance in the event of accidents, distress, or emergency, is not clear. It is true they will not be treated as diplomatic envoys with governmental immunity. If they are forced down in foreign territory or on the high seas they will have a temporary diplomatic immunity until they are returned to their own country. To what further protection are they entitled?

Seamen have long been regarded as wards of the admiralty. They are endowed special protection not accorded fellow workers on land, except by statute. Two ancient rights which they have acquired through practice and usage are the rights to maintenance and cure, and to a seaworthy vessel. Whenever a seaman becomes disabled during a sea voyage through sickness or accident, he is entitled to the continuance of his wages until restored to health, and to all medical expenses. This duty of the shipowner to provide all necessary treatment is an absolute one, not dependent on his negligence or on the freedom from fault of the seaman. The seaworthiness of the vessel is also an absolute requirement, and due diligence and care by the shipowner provides no exoneration for him if the court finds the vessel was not fit for whatever reason or cause. The concordant practice of maritime states has shaped a very strong maritime law in this regard.[31]

Will states whose nationals sojourn into space develop a similar set of protective rules for all spacefarers? I think it likely. Land-based rules for employer liability are shaped around work done "in the course of employment" and "within the scope of authority." On-duty and off-duty status and whether the aggrieved employee was deviating from his normal route or performance become important considerations. In space, as at sea, the spacefarer should be comprehensively protected without regard to any of the legal considerations. Whether his travel into space is for only a few days, or for weeks or months, he has a high risk occupation and his duty period, like that of the sailor, is for twenty-four hours each day. Consider how appropriate to the future spaceman is the language used by Justice Rutlege in describing the sailor's life. He wrote:

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[30]Outer Space Treaty, *supra* n. 9, art. 5.

[31]*See generally* on the subjects of maintenance and cure, and vessel seaworthiness, Gilmore and Black, The Law of Admiralty 151-155, 281-314 (2d ed. 1975).

> . . . . Unlike men employed in service on land, the seaman, when he finishes his day's work, is neither relieved of obligations to his employer nor wholly free to dispose of his leisure as he sees fit. Of necessity, during the voyage he must eat, drink, lodge and divert himself with the confines of the ship. In short, during the period of his tenure the vessel is not merely his place of employment; it is the framework of his existence. For that reason, among others, his employer's responsibility for maintenance and cure extends beyond injuries sustained because of, or while engaged in activities required by his employment.[32]

Federated States such as the U.S. should exercise legislative and judicial jurisdiction only at the national level. In the United States, a Federal Wrongful Death statute for outer space is needed which would preempt state law. It might be useful to extend the Death on the High Seas Act to activities in space.[33] The Federal Tort Claims Act should be amended to specifically cover acts in space and further revised to include claims based on strict liability for governmental conduct.[34] As long duration voyages and habitation in space become a reality, the relatively short statute of limitations for judicial recourse needs to be extended. In admiralty, by constitutional grant, Federal jurisdiction is exclusive (with certain exceptions not relevant here).[35] A uniform national space law is preferable to the hodgepodge of tort law applicable to aviation mishaps where state as well as Federal law is relevant. The working conditions of space crews and other personnel, minimum health standards for space flight, the provision for medical care and the extent of disciplinary authority of the mission and flight commanders must be defined only at the national and international levels.

On March 7, 1980, NASA promulgated rules on the authority of the shuttle commander in flight.[36] Shuttle commanders have been given authority similar to that held by sea captains and aircraft commanders including the power to arrest and to use force if necessary to maintain law and order in orbit. The rules also establish a chain of command on board the shuttle in the event the commander or his copilot are incapacitated. The new rules recognize that the shuttle will carry non-NASA scientists, engineers and foreign researchers.

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[32]*Waterman, S. S. Corp Jones,* 318 U.S. 724, 731, 733 (1943).

[33]Death on the High Seas Act, 41 Stat. 537, 46 U.S.C. § 7, 1-768 (1920). It has been judicially extended to death *over* the high seas. In *D'Aleman v. Pan American World Airways,* 259 F 2d 493 (1958) Judge Moore wrote, "The law would indeed be static if a passenger on a ship were protected by the Act and another passenger in the identical location three thousand feet above in a plane were not." It is extremely doubtful whether courts will judicially extend that Act to deaths on board an orbiting spaceship. But there is a similar need in space to unify the law. Courts have clearly held that the need for uniformity at sea required the DOHSA to supersede any state wrongful death statute. *See Dugas v. National Aircraft Corp.,* 438 F. 2d, 1386 (1971).

[34]The applicability of the Federal Tort Claims Act to acts in outer space has not been examined by the courts. F.T.C.A. 28 U.S.C. 1346 (b), 2671. Legislation precludes its applicability in cases where admiralty jurisdiction applies (see § 741-752 of the Suits in Admiralty Act 46 USC 741 et. seq. Since the F.T.C.A. is basically territorial in scope, it will not, in my view, be held to apply to negligence which occurs in the outer space environment.

[35]U.S. Const. art. 3, § 2.

[36]45 Fed. Reg. 14845 (1980), 14 C.F.R. 1214.

As one of the two great spacefaring states in the world, U.S. law and practice on space welfare will set the mark for other nations to follow as they develop their own space transport systems either in concert with other states or individually.

Salvage

Some years ago, Mr. Page, the President of the General Electric Missiles and Space Division, predicted there would be a million satellites in earth orbit in the 1990's. As time slips by, this prediction seems much too optimistic. There are approximately 4,500 space objects orbiting earth at the present, about one forth of which are functioning.[37] Although he may have miscalculated, the problems implicit in his prediction will be generated. The U.N. Secretariat's Report states it will "almost certainly" be necessary in time to sweep non-geostationary orbits planned for extremely large structures, such as those planned for in space construction of solar power satellites.[38] It is also true that much of the artifacts brought into space will be salvageable. The Astronaut Agreement does not address the question of salvage. It does provide that, upon request, objects launched into space found outside the launching state shall be returned or held at that state's disposal. It also provides for the recoupment of expenses.[39]

At sea, every recovery of lost or abandoned property, or property in peril, is subject to a salvage claim. Though limited to the value of the property saved, the award can be far in excess of expenses. This depends on the hazards involved, the value of the property at risk, the value of the property saved, and other considerations.

There is a duty imposed on the master of every vessel to render all assistance in saving life at sea and there is no reward for life salvage. The salvage of property at sea is a voluntary act and here a generous reward is possible except for those public ships such as Coast Guard vessels which have a duty imposed by law.[40]

The law of maritime salvage has been stated to be a part of international law, and the courts of maritime states have taken a liberal attitude toward entertaining salvage claims against a ship libeled in their ports. This is true even though the only contact with the port state is that the rescued ship was brought there after rescue. In maritime practice a wide variety of acts have been held to constitute salvage from simple towing to a place of safety to raising a sunken vessel. Will this body of maritime law have any relevance in space? I think the analogy is appropriate and that the need for uniformity, a *jus gentium* in space, is equally necessary. The question of who may become salvors, how the award should be computed and distributed, who is liable for the award, and what types of salvage may arise under contract will become important considerations in rescue operations in space.

It is not too early for an appropriate body such as COPUOS to consider whether a salvage law for space will require an international agreement comparable, perhaps, to

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[37]Interview with Dr. Perek, *supra* n. 11.

[38]*Supra* n. 6, at 18.

[39]Rescue and Return Agreement, art. 5 *supra* n. 9.

[40]*See* Norris, The Law of Salvage, § 172-179 (1958).

the Brussels Salvage Convention of 1910, or whether it can best develop through national law and practice.

Protection of the Environment

Already mentioned in connection with the need to control earth-threatening satellites such as unstable nuclear powered satellites is the need for a safety regime in space. However, the question of back and forward contamination in space is much broader than just the exercise of any right of removal of deleterious space objects. A special set of safety standards, uniformly adopted throughout the community of spacefaring states, is extremely desirable.

The Geneva High Seas Convention provides that every state shall draw up regulations to prevent pollution of the seas by the discharge of oil from ships and shall take measures to prevent pollution from the dumping of radio-active waste.[41] The International Maritime Consultative Organization (IMCO) has sponsored both the Convention for Oil Pollution Damage and the Convention on the Prevention of Marine Pollution by Dumping Wastes and Other Matters. The U.S. is a party to both.[42]

Respected admiralty lawyers have stated that ocean contamination presents "problems of staggering complexity as to which no consensus has yet emerged."[43] No less than 45 articles are devoted to the protection and preservation of the marine environment in the most recent draft of the Law of the Sea negotiating text. While it should be superficially attractive to transfer concepts being developed in this most vital area from the sea to space, it probably is not a useful analogy. First, because the law of the sea is itself in a state of confusion on this subject, and second, because the distinctive characteristics of the two environments call for entirely different approaches to the problem. Nevertheless, there must be a uniformity of approach, common standards and an international repository for the communication and exchange of all information on space contamination.

Conclusion

While the law of the sea does not represent an infallible guide to the developing law of space, space lawyers need some appreciation of maritime law. They need to become conversant with basic admiralty rules to vessel identification, navigation, management of ships, rights of seamen, the law of salvage, limitations of liability and governmental activity in sea commerce. It is with this perspective that they can compare the two environments, consider the relevance of maritime law, and select for adoption for space transport those legal principles which have proved beneficial to commerce on the oceans.

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[41]High Seas Convention, *supra* n. 22, art. 24, 25.

[42]Oil Pollution Convention, May 12, 1954 (1954) T.I.A.S. 4900; Dumping of Wastes Convention, December 29, 1972, T.I.A.S. 8165.

[43]*Supra* n. 30, at 826.